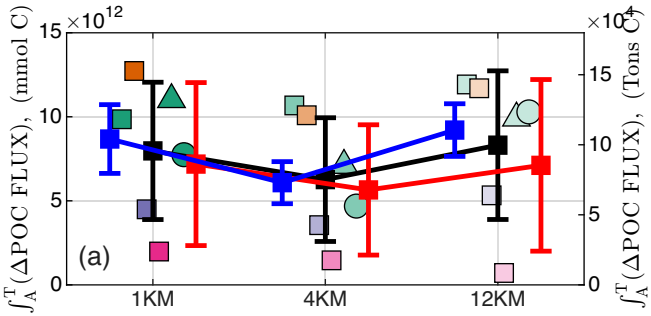
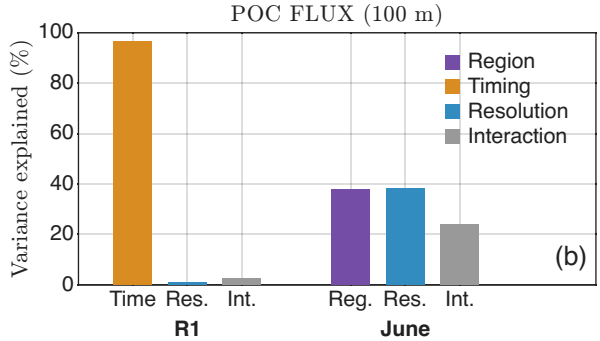
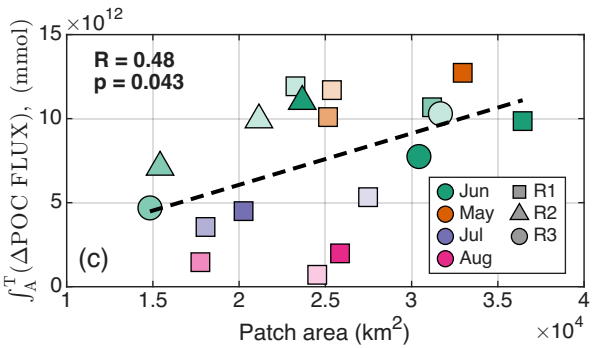
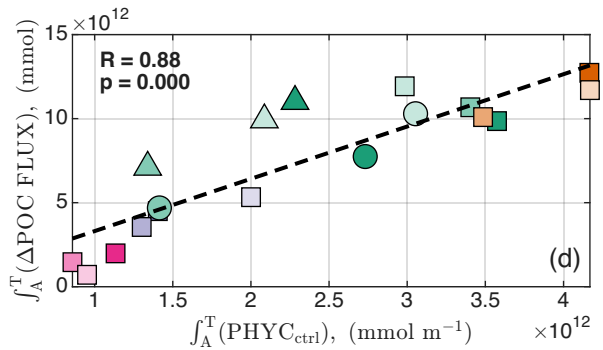
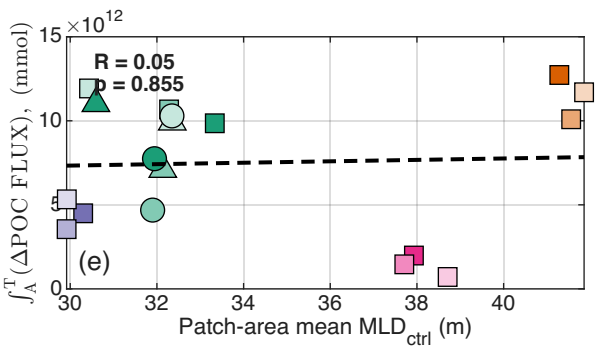
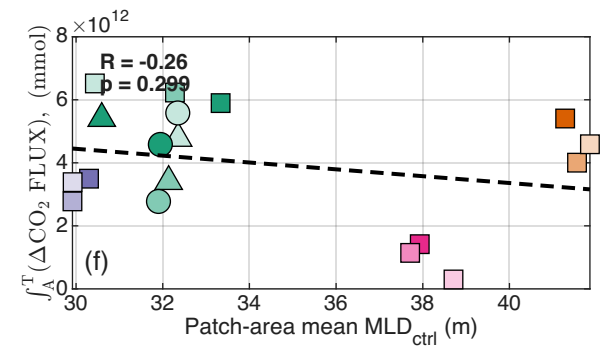

(a)
1KM
4KM
12KM

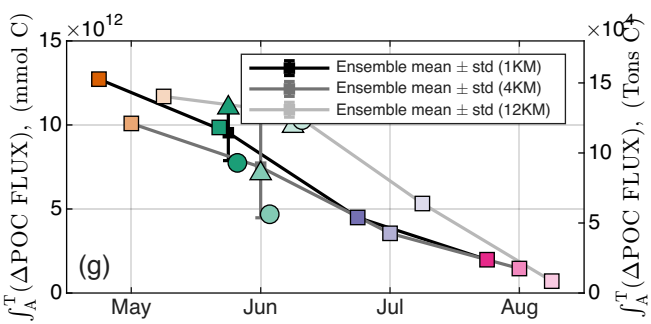

POC FLUX (100 m)
Region
Timing
Resolution
Interaction
Variance explained (%)
Time
Res.
Int.
Reg.
Res.
Int.
R1
June
(b)

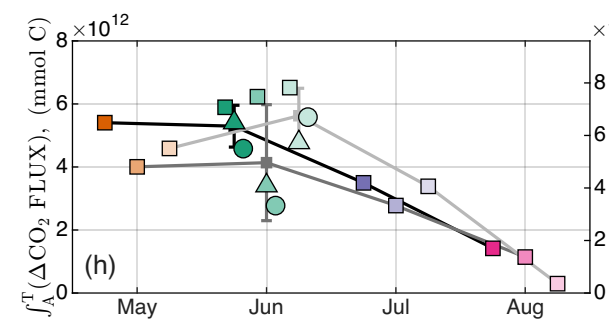

R = 0.48
p = 0.043
Jun
May
Jul
Aug
R1
R2
R3
(c)
Patch area (km²)


R = 0.88
p = 0.000
(d)


R = 0.05
p = 0.855
(e)
Patch-area mean MLD_ctrl (m)


R = -0.26
p = 0.299
(f)
Patch-area mean MLD_ctrl (m)


Ensemble mean ± std (1KM)
Ensemble mean ± std (4KM)
Ensemble mean ± std (12KM)
(g)
May
Jun
Jul
Aug


(h)
May
Jun
Jul
Aug